

1) Industrial Power Pack



Description:

Industrial power packs are engineered for continuous, high-load, and heavy-duty operation in manufacturing plants, process industries, and other demanding industrial environments. These systems serve as a centralized source of hydraulic energy, converting electrical or mechanical power into controlled hydraulic pressure and flow to operate various machines and equipment.

Typically, an industrial power pack consists of a hydraulic pump, electric motor or engine, reservoir, filtration system, valves, manifolds, cooling arrangements, and safety components. Each element is carefully designed to ensure stable performance, minimal downtime, and long service life, even under round-the-clock operation. The pump and motor are selected to deliver consistent flow rates and pressure levels, while robust reservoirs and cooling systems help manage heat generation during prolonged use.

Industrial power packs are widely used in applications such as presses, injection molding machines, material handling systems, machine tools, automation lines, and process equipment where precise and reliable hydraulic power is critical. They are capable of maintaining steady pressure and flow, which is essential for achieving accurate motion control, uniform force output, and repeatable performance.

In addition, these power packs are often customized to meet specific operational requirements, including pressure ratings, flow capacities, control logic, and environmental conditions. Features such as pressure relief valves, accumulators, advanced filtration, and monitoring instruments enhance safety, efficiency, and system reliability. Overall, industrial power packs play a vital role in ensuring smooth, efficient, and uninterrupted operation of hydraulic systems in demanding industrial applications.

Main Components:

- AC Electric Motor
- Hydraulic Pump (Gear / Vane / Piston)
- Oil Reservoir
- Directional Control Valves
- Pressure Relief and Control Valves
- Filters and Cooling System

Applications:

- Hydraulic Presses
- Material Handling Systems
- Machine Tools
- Steel and Cement Plants

Advantages:

- Heavy-duty construction
- Suitable for continuous operation
- Customizable pressure and flow

2) AC/DC Power Pack



Description:

AC/DC power packs are compact hydraulic power units designed to operate on either alternating current (AC) or direct current (DC) electrical supply. This dual-power capability makes them highly versatile and suitable for applications where a conventional power source may not always be available. Typically, AC operation is used in fixed or workshop environments, while DC operation—often through batteries—is preferred in mobile, remote, or emergency situations.

These power packs consist of key components such as an electric motor (AC or DC), hydraulic pump, oil reservoir, control valves, pressure relief valves, and electrical controls. Despite their compact size, they are engineered to deliver sufficient pressure and flow to operate hydraulic cylinders, motors, and other actuators efficiently. Their design emphasizes portability, ease of installation, and quick response time.

AC/DC power packs are widely used in mobile equipment like tail lifts, scissor lifts, hydraulic jacks, recovery vehicles, and portable lifting systems. In emergency applications, they play a critical role in rescue tools, backup hydraulic systems, and equipment that must function reliably during power outages or in field conditions. The ability to switch between AC and DC ensures uninterrupted operation and enhances system reliability.

Additional advantages include low maintenance requirements, energy-efficient operation, and customizable configurations to suit specific pressure, flow, and control needs. Due to their flexibility, compact construction, and dependable performance, AC/DC power packs are an essential solution for modern hydraulic systems that demand mobility, reliability, and adaptability across various operating conditions.

Main Components:

- AC or DC Motor (12V / 24V)
- Compact Hydraulic Pump
- Small Oil Tank
- Solenoid Valves
- Relief and Check Valves

Applications:

- Hydraulic Lifts
- Scissor Lifts
- Dock Levelers
- Tippers and Trailers

Advantages:

- Compact size
- Suitable for mobile use
- Easy installation